

Phase 1.5D — Categorical Encoding Comparison

Experiment: E1 vs. encoding variants | **Dataset:** IEEE-CIS (test set) **Status:** E1 (V1) is IMMUTABLE — comparison only, not overwrite

E1b Decision Criterion

A variant is recommended for E1b only if it shows **consistent improvement across**:

- PR-AUC (test)
- Recall @ 1% FPR (test)
- Precision @ Top-1,000 (test)
- No significant increase in training cost

A single-metric marginal gain does NOT justify E1b.

Multi-Metric Comparison Table

Metric	V1 LabelEncoder (E1)	V2 Native Categorical	V3 Frequency Encoding
PR-AUC (test)	0.5362	0.5143	0.5311
ROC-AUC (test)	0.9034	0.8940	0.9029
Recall (test, frozen threshold)	0.4427	0.4405	0.4874
Precision (test)	0.6269	0.5935	0.5428
F1 (test)	0.5189	0.5057	0.5136
Recall @ 1% FPR (test)	0.4445	0.4322	0.4477
Precision @ Top-1,000 (test)	0.8260	0.8100	0.8440
PR-AUC (validation)	0.5893	0.5772	0.5754
Training time (seconds)	76.73	42.88	40.39
Number of trees	980	744	747
Frozen threshold	0.668643	0.673032	0.666350

E1b Recommendation

Decision: E1b is NOT justified. E1 (V1 LabelEncoder) remains the final baseline.

Decision	Reasoning
E1b justified?	NO — V1 outperforms V2 and V3 on all primary metrics

Decision	Reasoning
Recommended encoding	V1 LabelEncoder (current E1)
Next step	Keep E1 as the immutable baseline. Proceed to Phase 2 GRU.

Detailed reasoning:

- **V2 (Native Categorical)** is strictly worse than V1 on every metric: PR-AUC drops from 0.5362 → 0.5143 (−0.022), Recall@1%FPR drops from 0.4445 → 0.4322, P@Top-1K drops from 0.8260 → 0.8100. Despite using LightGBM's theoretically superior Fisher-split algorithm, V2 underperforms — likely because LightGBM's tree splits already compensate for LabelEncoder's ordinal ordering at this data scale.
- **V3 (Frequency Encoding)** is slightly better than V2 but still worse than V1 on PR-AUC (0.5311 vs 0.5362) and Recall@1%FPR (0.4477 vs 0.4445). Its P@Top-1K (0.8440) marginally exceeds V1 (0.8260), but this single-metric edge does not satisfy the multi-metric consistency criterion for E1b.
- **V1 trains in 76.7s** (more trees due to no early stopping), V2/V3 train faster (~42s) but with fewer trees and lower scores.
- No variant shows consistent cross-metric improvement. **E1b is not warranted.**

Notes on Encoding Strategies

- V1 — LabelEncoder:** Assigns integers 0, 1, 2, ... to categories in sorted order. Creates artificial ordinal relationships. However, LightGBM's tree-splitting can partially compensate by treating each split as binary.
- V2 — Native Categorical:** LightGBM internally uses the Fisher optimal split algorithm for categorical features, which finds the optimal subset partition rather than imposing ordinal order. Generally preferred for high-cardinality categoricals.
- V3 — Frequency Encoding:** Replaces each category with its training-set count. Encodes popularity/rarity information. No target information used (unlike target encoding). Works well when frequency is a meaningful signal (e.g., rare email domains = higher fraud risk).

Generated by: src/audit/encoding_comparison.py | Phase 1.5D | E1 IMMUTABLE